

MPLADS Cybersecurity & Audit Engine: Complete Architectural Overhaul Specification

Smart India Hackathon (SIH 2026) — Problem Statement 26102

Prepared For: Project Development Team (All Domains)	Domain Scope: ML Engine, Backend, Data Pipeline, Frontend, DevOps	Reference Policy: MPLADS Guidelines 2023 & eSAKSHI Model 1A TSA Framework
---	--	--

1. Executive Summary & Context of Realignment

A critical review of the actual operational mechanics of the **Members of Parliament Local Area Development Scheme (MPLADS)** revealed a fundamental divergence between standard financial fraud assumptions and actual statutory governance under the **MPLADS Guidelines 2023** and the **2025 Model 1A TSA Hybrid payment architecture**.

Initial versions assumed a decentralized, physical cash-transfer model where direct budget overdrafts (expenditure > sanctioned) were the primary risk vector. In reality, modern MPLADS operations are anchored on the **eSAKSHI portal** integrated with the **Public Financial Management System (PFMS)** directly through **RBI Assignment Accounts**. Physical fund parking at district levels is eradicated; funds release just-in-time upon invoice approval. Consequently, actual systemic vulnerabilities stem from **SLA sanction bottlenecks, Third-Party Inspection (TPI) evasion, election-cycle spending surges, demographic quota violations, and discrepancies between central eSAKSHI records and district RTI edge records**.

2. Cross-Domain Comparative Matrix: Baseline vs. Overhaul

Domain	Previous Baseline / Legacy Assumption	Required Overhaul (MPLADS Research Compliant)	Priority
ML Engine (AI/Risk)	4-feature Isolation Forest (sanctioned, expenditure, progress, burn_rate). Assumes simple budget burn rate anomalies.	8-Dimensional Statutory Risk Vector integrating 45-day Sanction SLAs, 1-year execution limits, 18-month MP demitting cutoffs, 18% GST deductions, TPI compliance, and election cycle surges.	CRITICAL
Data Pipeline (ETL)	Generic single CSV cleaning script. Drops missing IDs, fills nulls with zero.	Bifurcated Ingestion Engine: (1) eSAKSHI Central Digital Node (Post-2023 JSON/CSV) & (2) Physical District Edge Node (RTI OCR digitization). Distinguishes MPLADS from MLALADS.	CRITICAL
Backend Data API	Basic CRUD operations for works and anomalies. Generic financial models.	Statutory Schema Expansion: Adds lifecycle milestones, TPI status, geo-tagging flags, SC/ST demographic flags, and Model 1A TSA payment transaction states.	MAJOR
Backend Case Mgmt	Static case review table without role mapping.	Role-Based Administrative Routing: Directs SLA delays to IDAs, QA deficits to IAs/TPI officers, and demographic non-compliance to SNAs. Includes automated RTI petition generator.	MAJOR
Frontend Dashboard	Mock KPI cards and simple tables showing generic risk scores.	Civic Audit Intelligence UI: Statutory SLA countdowns, 15% SC / 7.5% ST demographic gauges, TPI quality assurance badges, pre-election surge timelines, and interactive RTI builder.	MAJOR
Cloud & DevOps	Independent container definitions with placeholder networking.	Unified Microservice Gateway: Docker Compose orchestration with automated healthchecks, CORS hardening, shared volumes, and Nginx reverse proxy.	NORMAL

3. Comprehensive Domain-by-Domain Action Plan

Domain 1: ML Engine (`ml-engine/`)

The ML Engine must transition from a simplistic numerical outlier detector into a **Hybrid Statutory Risk & Anomaly Engine**.

- **18% GST Normalization:** Civil construction contracts under MPLADS incur 18% GST (absorbing ~₹90 lakh per ₹5 crore allocation). The feature engine must normalize net asset creation expenditure against gross sanctions ($\text{expenditure_net} = \text{expenditure} * 0.82$) to prevent false-positive burn rate alarms.
- **8-Dimensional Feature Space:**

```
features = [  
    'sanction_delay_days',      # Days from recommendation to IDA sanction (SLA: <= 45 days)  
    'execution_delay_ratio',    # Days elapsed / Sanctioned timeline (SLA: <= 365 days)  
    'post_demit_overdue_flag',  # 1 if project active > 18 months post-MP demit, else 0  
    'gst_adjusted_burn_rate',    # (expenditure * 0.82) / (progress_percent + 1.0)  
    'tpi_compliance_flag',      # 0 if Third-Party Inspection missing, 1 if verified  
    'geotag_photo_flag',        # 0 if geo-tagged completion photo missing, 1 if verified  
    'election_proximity_index',  # Proximity in months to general election (detects rush  
    spending)  
    'sc_st_quota_deviation'     # Deviation from 15% SC / 7.5% ST statutory targets  
]
```

- **Statutory Explainability Engine (`app/explain.py`):** Replace generic strings with Paragraph-specific legal and policy audit citations:
 - *Paragraph 3.2.4 SLA Alert:* "Implementing District Authority exceeded the statutory 45-day sanction limit."
 - *Paragraph 4.7 QA Alert:* "Disbursement executed without mandatory Third-Party Inspection (TPI) certification."
 - *Paragraph 10.6.1 Tenure Alert:* "Work uncompleted > 18 months after recommending MP demitted office."
 - *Paragraph 3.2.12 Delay Alert:* "Civil execution exceeded standard 1-year timeline without recorded terrain justification."
- **Model Retraining & Evaluation (`training/`):** Re-train Isolation Forest + rule blend on synthesized and historical audit records, generating benchmark metrics matching CAG/NABCONS audit patterns.

Domain 2: Data Pipeline & Ingestion (`data-pipeline/`)

- **Bifurcated Sourcing Architecture:**
 1. **Central Node:** Ingest standardized records from the eSAKSHI portal (`mplads.mospi.gov.in`) using modern work codes (MPL-XX-NN/YYYY-YY).
 2. **Edge Node (RTI Extraction):** Support localized data ingestion (OCR / structured CSV) for physical records obtained from District Magistrate MPLADS cells under Section 7(1) of the RTI Act.

- **Strict MPLADS vs. MLALADS Separation:** Prevent spatial conflation in multi-tier constituencies where state MLAs and federal MPs allocate funds in overlapping geographical coordinates.
- **Data Harmonization (`clean.py`):** Parse multi-format dates (YYYY-MM-DD, DD-MM-YYYY), validate constituency and state mapping, and flag multi-hop transaction discrepancies.

Domain 3: Backend Data API (`backend-data-api/`)

- **Database Schema Migration (`app/models/`):**
 - Work model: Add recommendation_date, sanction_date, target_completion_date, actual_completion_date, tpi_status, geotag_photo_url, and demographic_category.
 - Transaction model: Represent Model 1A TSA stages (demand_generated, pfms_validated, rbi_transferred, disbursed_to_vendor).
 - FlaggedAnomaly model: Store categorized audit violation codes and AI confidence scores.
- **Advanced Filtering Routes (`app/routes/`):** Expose query parameters for filtering works by SLA sanction status, inspection completion, SC/ST demographic allocations, and high-risk vendor concentration.

Domain 4: Backend Case Management (``backend-case-management/``)

- **Administrative Workflow & Escalation:**
 - Automatically create case files when project risk exceeds threshold (`risk_score >= 70`).
 - Assign cases to appropriate administrative tiers: **Implementing District Authority (IDA)** for SLA bottlenecks, **Implementing Agency (IA)** for missing site photos/TPI, and **State Nodal Authority (SNA)** for quota deficits.
- **Automated RTI Letter Generator:**
 - Generate pre-filled, legally formatted RTI application letters addressed to the Public Information Officer (PIO) of the District Magistrate's office for edge-level verification of contractor bills and TPI inspection reports.

Domain 5: Frontend Dashboard (``frontend-dashboard/``)

- **Statutory SLA Tracking Dashboard:** Visual KPI widgets showing:
 - Sanction compliance rate (% works sanctioned within 45 days).
 - 1-year execution adherence (% works completed within 365 days).
 - Post-demit resolution (% legacy works completed within 18 months).
- **Demographic Equity & Quota Gauges:** Real-time progress bars showing compliance with mandatory 15% SC (₹75L) and 7.5% ST (₹37.5L) allocations per MP per financial year.
- **Quality Assurance & TPI Verification Badges:** Explicit visual tags on project cards indicating Third-Party Inspection status, geo-tagged photograph uploads, and Utilization Certificate (UC) issuance.
- **Political Business Cycle Analytics:** Historical spending distribution charts highlighting disproportionate expenditure surges in Year 4/5 of an MP's 5-year tenure.
- **Auditor Explainability Drawer:** Clean UI drawer displaying detailed CAG/eSAKSHI statutory violation citations alongside AI risk heatmaps.

Domain 6: Cloud DevOps & System Integration (``cloud-devops/``, ``docker-compose.yml``)

- **Service Interoperability:** Ensure unified Docker Compose network bridging data-pipeline, backend-data-api, backend-case-management, ml-engine, and frontend-dashboard.
- **Resilient Microservice Contracts:** Maintain backwards-compatible Pydantic models with optional field defaults so frontend and backend services communicate reliably during incremental rollout.

4. Implementation Timeline & Milestone Roadmap

Phase	Target Domain	Key Deliverables	Estimated Effort
Phase 1	ML Engine & DB Schema	Implement 8D feature engineering, update <code>predict.py</code> , enhance <code>explain.py</code> with 2023 Guidelines citations, and expand SQLAlchemy models.	2 Days
Phase 2	Data Pipeline & ETL	Refactor <code>clean.py</code> to parse statutory dates, TPI flags, SC/ST classifications, and build edge RTI ingestion parser.	2 Days
Phase 3	Backend APIs & Case Mgmt	Update REST endpoints, implement role-based case escalation workflows, and build automated RTI draft petition generator.	2 Days
Phase 4	Frontend UI Realignment	Build SLA compliance widgets, SC/ST equity gauges, TPI status badges, pre-election trend visualizations, and explainability modal.	3 Days
Phase 5	DevOps & End-to-End Testing	Perform full-stack Docker Compose smoke tests, CI/CD validation, and end-to-end hackathon demo verification.	1 Day

Summary for Hackathon Jury & Technical Auditors: By shifting the architecture from generic financial heuristics to the official **MPLADS 2023 Guidelines & eSAKSHI Model 1A TSA framework**, this project provides a government-grade, legally compliant, and actionable civic auditing platform that directly addresses **SIH 2026 Problem Statement 26102**.